IOT & GAME DEVELOPMENT PROJECT

NeuroDrive: Hardware-Controlled Web Racing System

An ESP32 & MPU6050 Motion-Sensing Browser Game Architecture

MICROCONTROLLER	SENSORS	COMMUNICATION
ESP32-WROOM-32	MPU6050 (6-DOF IMU)	Wi-Fi AP / HTTP JSON API

1. Project Overview

NeuroDrive is an integrated hardware-software system that transforms physical motion into real-time interactive browser gameplay. Powered by an **ESP32** microcontroller and an **MPU6050 6-Axis Inertial Measurement Unit (IMU)**, the controller hosts its own Wi-Fi Access Point and Web Server. Players tilt the handheld physical sensor to steer, accelerate, and brake inside an HTML5 Canvas/CSS racing game running on any connected mobile device or computer.

20 Hz	α = 0.35	115.2 kbps	80
POLLING RATE (50MS)	EMA FILTER FACTOR	BAUD RATE	HTTP WEB SERVER PORT

2. Hardware & Interfacing Architecture

The system utilizes the ESP32 to communicate with the MPU6050 accelerometer and gyroscope chip via the I2C (Inter-Integrated Circuit) bus protocol.

HARDWARE COMPONENTS

- **ESP32 Development Board:** Dual-core LX6 microcontroller handling network traffic and sensor reads.
- **MPU6050 Module:** Contains 3-axis accelerometer and 3-axis gyroscope with 16-bit ADC output.
- **Power Source:** Micro-USB / LiPo Battery supply.

I2C PIN CONNECTIONS

MPU6050 Pin	ESP32 GPIO	Description
VCC	3.3V / 5V	Power Supply
GND	GND	Common Ground
SDA	GPIO 19	I2C Data Line
SCL	GPIO 18	I2C Clock Line

3. Sensor Data Processing & Filtering

Raw sensor readings from IMUs suffer from mechanical noise, high-frequency vibration, and electronic jitter. To ensure smooth steering and speed control without latency, the ESP32 firmware implements an **Exponential Moving Average (EMA)** low-pass filter.

Filter Equation: $y_t = \alpha \cdot x_t + (1 - \alpha) \cdot y_{t-1}$

Where $\alpha = 0.35$, x_t is current raw sensor value, and y_{t-1} is previous filtered value.

The MPU6050 register memory map is read via a 14-byte burst read at address $0x3B$, retrieving 6-DOF data:

- **Gyroscope X-Axis (gx):** Filtered with EMA ($\alpha = 0.35$) to control steering tilt response.
- **Accelerometer Y-Axis (ay):** Filtered with EMA to control pitch (acceleration/braking).
- **Accelerometer & Gyroscope Z/Y Axes:** Passed for secondary telemetry.

4. Software Architecture & Network Flow

The firmware sets up an Access Point named `MPU6050_Controller` with IP `192.168.4.1`. The HTTP server routes requests as follows:

Endpoint	HTTP Method	Content-Type	Function & Description
`/`	GET	`text/html`	Serves full interactive single-page JavaScript game UI stored in PROGMEM/rawliteral.
`/data`	GET	`application/json`	Returns high-speed JSON payload with filtered IMU parameters. Header includes `Cache-Control: no-cache`.

Sample Telemetry Payload (JSON)


```
{
  "gx": 124.5, "gy": -12, "gz": 45,
  "ax": 1024,   "ay": -3420.8, "az": 15420
}
```


5. Game Engine & Control Dynamics

The frontend is written in vanilla JavaScript with HTML5 Canvas-like styled DOM elements. It implements a full game loop driven by `requestAnimationFrame`.

GAME MECHANICS

- **Steering Dynamics:** Angular tilt mapped relative to baseline with a deadzone of ± 400 units to eliminate drift.
- **Speed Dynamics:** Accelerometer pitch increases speed up to $12 + score/300$ units/sec or brakes when pulled back.

INTERACTIVE SYSTEMS

- **Auto-Calibration:** 1.2-second sample averaging at startup to lock zero-reference point.
- **Web Audio API:** Synthesizes real-time sound effects (beeps, powerups, crashes) without external media assets.

- **Dynamic Spawning:** Coins (+25 pts), Shields (temporary immunity), Rocks, Cones, and Rival Vehicles.

- **Signal Quality Indicator:** Live heartbeat detector flags signal loss if data fetch exceeds 700ms.

6. Technical Summary

NeuroDrive demonstrates a zero-latency, cross-platform hardware-to-browser controller pipeline. By utilizing internal Wi-Fi AP hosting, lightweight EMA signal conditioning, and REST-style polling, it bridges physical motion control with browser-based gaming without needing external internet access or dedicated mobile applications.